Exp 29 MATLAB CODE

```
clc;
clear;
close all;

% GSM TDMA Parameters
numUsers = 8;
numSlots = 8;
bitsPerSlot = 20;

% Generate Random Data for Each User
data = randi([0 1], numUsers, bitsPerSlot);

% TDMA Frame Generation
tdmaFrame = zeros(numSlots, bitsPerSlot);

for i = 1:numUsers
    tdmaFrame(i,:) = data(i,:);
end

% Display User Data
disp('Data of Each User:');
disp(data);

disp('TDMA Frame:');
disp(tdmaFrame);

% Serial Transmission
```

```
txSignal = reshape(tdmaFrame', 1, []);

% Add AWGN Noise
SNR = 10;
signalPower = mean(txSignal.^2);
noisePower = signalPower / (10^(SNR/10));
noise = sqrt(noisePower) * randn(size(txSignal));
rxSignal = txSignal + noise;

% TDMA Receiver
rxFrame = reshape(rxSignal, bitsPerSlot, numSlots)';

% Detect Received Data
detectedData = rxFrame > 0.5;

% BER Calculation
errors = sum(data(:) ~= detectedData(:));
BER = errors / numel(data);

% Display Results
fprintf('Number of Users = %d\n', numUsers);
fprintf('Time Slots per Frame = %d\n', numSlots);
fprintf('SNR = %d dB\n', SNR);
fprintf('Total Bits Transmitted = %d\n', numel(data));
fprintf('Bit Errors = %d\n', errors);
fprintf('Bit Error Rate = %.4f\n', BER);

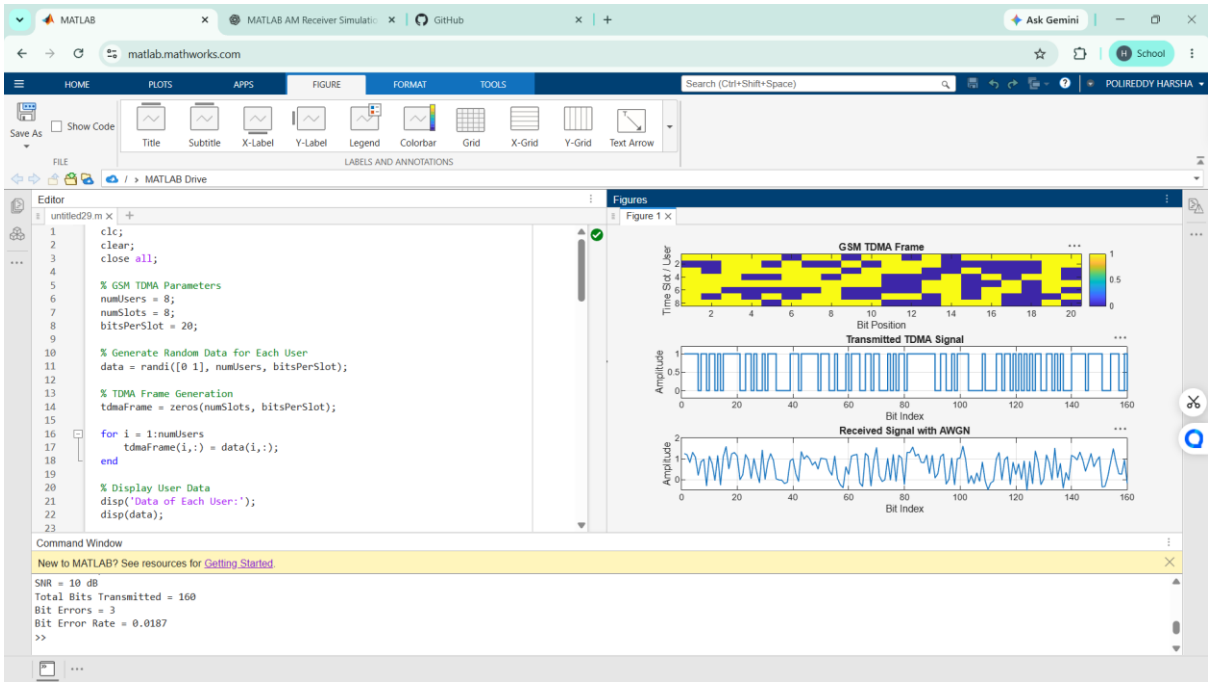
% Plot TDMA Frame
```

```
figure;

subplot(3,1,1);
imagesc(tdmaFrame);
title('GSM TDMA Frame');
xlabel('Bit Position');
ylabel('Time Slot / User');
colorbar;

% Plot Transmitted Signal
subplot(3,1,2);
stairs(txSignal(1:160), 'LineWidth', 1.2);
title('Transmitted TDMA Signal');
xlabel('Bit Index');
ylabel('Amplitude');
ylim([-0.2 1.2]);
grid on;

% Plot Received Signal
subplot(3,1,3);
plot(rxSignal(1:160), 'LineWidth', 1);
title('Received Signal with AWGN');
xlabel('Bit Index');
ylabel('Amplitude');
grid on;
```


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LABELS AND ANNOTATIONS
/ › MATLAB Drive
Editor
untitled29.m
1 clc;
2 clear;
3 close all;
4
5 % GSM TDMA Parameters
6 numUsers = 8;
7 numSlots = 8;
8 bitsPerSlot = 20;
9
10 % Generate Random Data for Each User
11 data = randi([0 1], numUsers, bitsPerSlot);
12
13 % TDMA Frame Generation
14 tdmaFrame = zeros(numSlots, bitsPerSlot);
15
16 for i = 1:numUsers
17 tdmaFrame(i,:) = data(i,:);
18 end
19
20 % Display User Data
21 disp('Data of Each User:');
22 disp(data);
23
Figures
Figure 1
GSM TDMA Frame
Time Slot / User
Bit Position
Transmitted TDMA Signal
Amplitude
Bit Index
Received Signal with AWGN
Amplitude
Bit Index
Command Window
New to MATLAB? See resources for Getting Started.
SNR = 10 dB
Total Bits Transmitted = 160
Bit Errors = 3
Bit Error Rate = 0.0187
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